

MIDTERM 2—GRADING RUBRIC FOR SECTIONS 2 AND 3

ECON101—INTRODUCTION TO MICROECONOMICS.

FALL 2024

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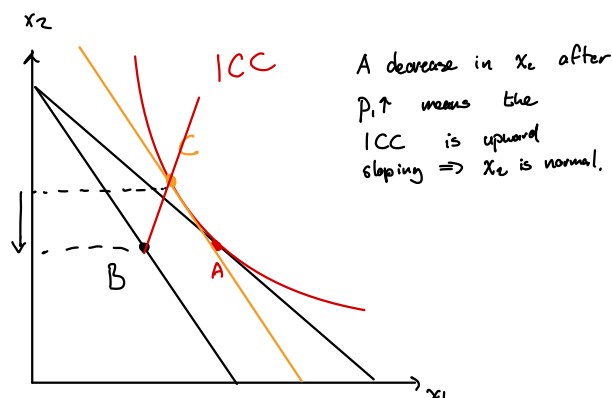
NAZARBAYEV UNIVERSITY

Part II: True/False Questions

For each of the following statements, indicate whether it is True or False, and provide a justification for your answer. Points will be awarded based on your explanation, not simply on choosing True or False. Without a justification, even a correct True/False answer will receive no points. Please write your answer only in the space provided. Each item is worth 10 points, for a total of 40.

1. Medina consumes only two goods, x_1 , and x_2 . When the price of x_1 increases, she consumes less of good x_2 . Claim: x_2 can't be a Giffen good.

- TRUE.
- When p_1 increases the substitution effect leads to more consumption of x_2 , so the income effect must be negative if the total on x_2 is negative.
- This implies that x_2 is a normal good.
- Normal goods can't be Giffen, because they need to be inferior.
- See the graph below for a graphical explanation. (Graph not needed for full points.)
- The argument is constructed via the points above. Award points based on which part of the argument they had.



2. Let F denote food from GreenFood, and C denote consumption on everything else. Suppose that the typical NU student has a yearly budget of ₸400,000; the typical student spends ₸40,000 on F and ₸360,000 on C . NU's administration is planning to assist students by considering giving either an extra stipend in cash of ₸80,000 or giving students coupons for GreenFood worth ₸80,000.

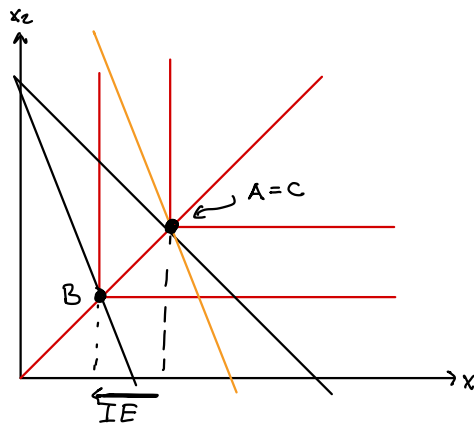
(a) If F is a normal good, then a student will be strictly better off with ₸80,000 in cash than with ₸80,000 of coupons.

- FALSE

- If the average students want to spend more than 80,000 in GreenFood when she receives the cash transfer, then she can also buy the same bundle when she receives 80,000 worth of coupons.
 - Therefore, she could be indifferent between both programs.
 - The argument is constructed via the points above. Award points based on which part of the argument they had.
- (b) If F is an inferior good, then a student will be strictly better off with $\bar{T}80,000$ in cash than with $\bar{T}80,000$ of coupons.
- TRUE
 - Since GreenFood is inferior, we know that when receiving 80,000 in cash, she will spend less than 40,000 in GreenFood. However, this bundle is not affordable when she receives 80,000 in coupons.
 - Therefore, the student will have to choose another bundle at a lower indifference curve.
 - The argument is constructed via the points above. Award points based on which part of the argument they had.

3. Rakhim consumes only sodas x_1 and pizza x_2 . Suppose that his preferences can be represented by $U(x_1, x_2) = \min\{x_1, x_2\}$, so sodas and pizza are perfect complements: Rakhim wants to drink one soda for every pizza he eats. If the price of soda increases, Rakhim's demand for both sodas and pizzas will decrease, and all change in the demand will be due to the substitution effect.

- FALSE.
- All change in demand comes from the income effect.
- See the graph below for reference. (Graph not needed for full points.)



Part III: Long Problem on Consumer Theory

Answer each question only in the space provided. Be organized, clear, and concise. Points will be deducted for lack of organization or if your answer is unclear. The point value for each item is indicated in parentheses. This section is worth a total of 56 points.

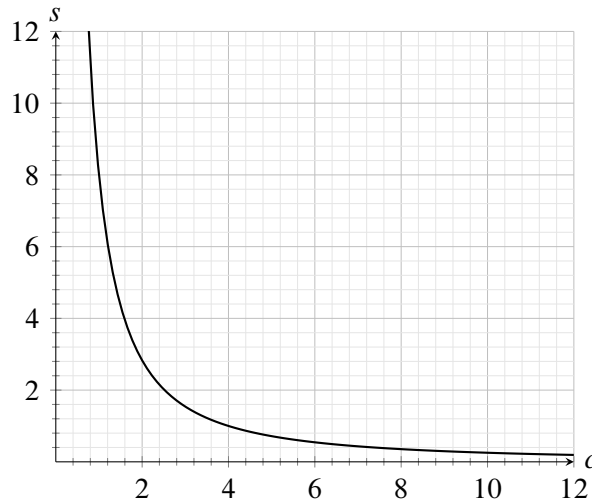
Nazerke likes eating sweet pastries s and drinking coffee c for breakfast. Her preferences can be represented by the utility function

$$U(c, s) = c^3 s^2.$$

For your reference $MU_s = 2c^3 s$, and $MU_c = 3c^2 s^2$.

1. (4 points) Graph Nazerke's indifference curve corresponding to utility level $U = 64$. Coffee should be on the horizontal axis. To get the equation: $64 = c^3 s^2$. Taking the only the positive square root leads to

$$s = \frac{8}{c^{3/2}}$$



- A convex, moderately accurate graph is enough for full credits. Axes should be *correctly labeled*, c for the horizontal, s for the vertical, for full points.
 - If axes are not labeled, deduct 1 point.
 - The equation for the indifference curve doesn't need to be there, but you can give 2 points as partial credit if it is there but the graph is not.
2. Suppose that Nazerke budgets m tenges for breakfast every week, and the price of coffee and sweet pastries are p_c and p_s respectively.
- (a) (4 points) Write down the problem that Nazerke solves when she is deciding how many coffees and sweet pastries to consume each week. You are only required to write the problem, not solve it!

$$\begin{aligned} \max_{c,s} \quad & c^3 s^2 \\ \text{subject to} \quad & p_c c + p_s s = m \end{aligned}$$

- Full points is the problem above. The budget restriction can appear with inequality.
 - If only the budget restriction appears, 1 point.
 - If the budget restriction and the marginal rate of substitution equal the price ratio appear in here, 2 points.
 - If the MRS is the only thing that appears 1 point.
- (b) (4 points) Solve for how many coffee and pastries Nazerke would consume as a function of p_c , p_s , and m .
The absolute value of the MRS is:

$$|MRS| = \frac{MU_c}{MU_s} = \frac{3c^2 s^2}{2c^3 s} = \frac{3s}{2c}$$

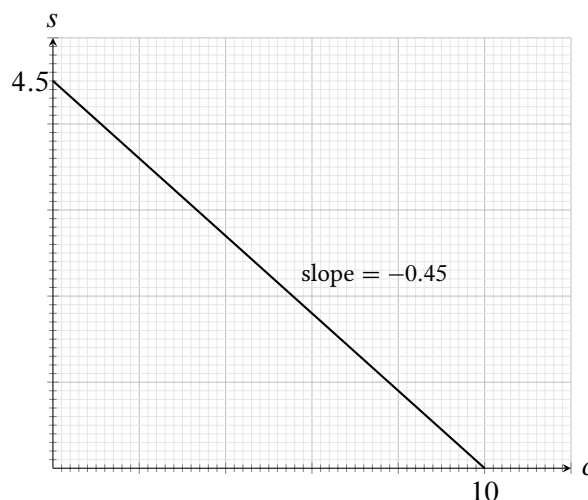
The optimal bundle satisfies:

$$\frac{3}{2} \frac{s}{c} = \frac{p_c}{p_s}, \quad p_c c + p_s s = m$$

Solving the two equations together yields the Marshallian demands (which are functions of prices and income):

$$c^M = \frac{3}{5} \frac{m}{p_c} \quad s^M = \frac{2}{5} \frac{m}{p_s}$$

- Some sort of the above needs to appear, *together* with the correct demand expressions for full points.
 - If only the correct demand expressions appear, 3 points.
 - If only the MRS equal to price ratio condition appears, 1 point.
 - If only the budget constraint appears, 1 point.
 - If both the MRS equal to price ratio and the budget line appears, and some attempt was made to solve them, 2.5 points.
- (c) (4 points) Are coffee and pastries normal goods or inferior goods? Why?
- Both goods are normal because the demand increases positively with Income, as can be seen from the demand expressions above.
- Full credit for the argument needs to allude to the demand equations. They need to recognize that the demand increases with income for the expression for demand that they got in the previous item. Their demand expression might be incorrect in the previous item, but if they recognized normality by alluding to their expression, then it is okay.
 - Any deviation from the previous explanation gives no points. I mentioned it since the last midterm: arguments need to be supported by correct reasoning and/or correct mathematics.
3. Now suppose that Nazerke budgets ₹4500 for breakfast every day and only eats at Daily Cup, where the price of a pastry is ₹1000 and the price of a cup of coffee is ₹900. However, because Nazerke is part of the loyalty program at Daily cup, she receives a 50% discount on coffee.
- (a) (4 points) Graph Nazerke's budget with coffee in the horizontal axis.

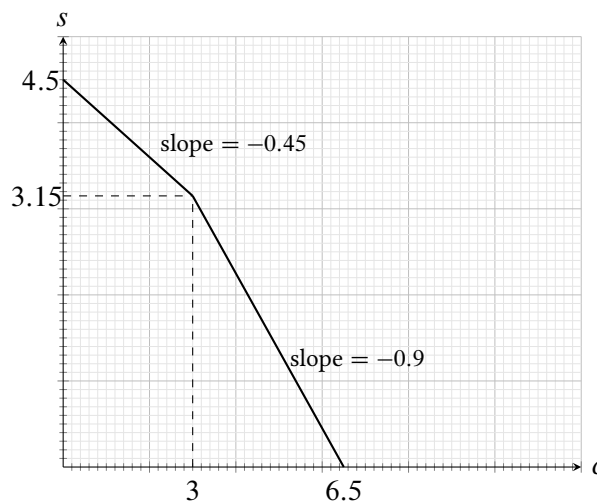


- Full credits are for graph like above, with correctly labeled axes and intercepts. The slope can be missing if they wrote the intercepts correctly.
 - If axes are not marked, but intercepts are, deduct 1 point.
 - If intercepts are not marked, but axes are, deduct 2 points.
 - If they wrote the equation of the budget line: $450c + 1000s = 4500$ or $s = 4.5 - 0.45c$, but didn't graph it, give 2 points.
 - The correct equation of the budget line plus an incorrect graph is 3 points.
- (b) (4 points) Find the number of coffee and pastries that Nazerke chooses to buy.

$$c^M = \frac{3}{5} \times \frac{4500}{450} = 6, \quad s^M = \frac{2}{5} \times \frac{4500}{1000} = 1.8.$$

- This was just substituting values into the demand functions that they should have found previously in 2.b.
 - In case they didn't find it, they should follow the standard procedure of equating the *MRS* to the appropriate price ratio of 0.45 and solve it together with the budget line.
 - Full credits are for getting the right answer. A maximum of two points of partial credit can be awarded if they try some *coherent* procedure, but didn't reach the solution.
 - Note that several students might have missed the traditional assumption that we always make in the problem sets and the lectures: *fractional amounts are possible*. If any student rounded the consumption of pastries to 2 that is incorrect because it is not feasible. If they rounded it down to 1, deduct only one point if the procedure is correct.
4. After a while, Daily cup decides to cancel the previous loyalty program and changes it to a new one: now a customer only gets a 50% discount on the first three cups of coffee they buy in any given day.

- (a) (5 points) Graph Nazerke's new budget set. Coffee should go on the horizontal axis. Carefully label every important point on the graph.



- Full credits are for a graph like the one above, with correctly labeled axes and intercepts. The slopes can be missing if they wrote all the relevant points and intercepts correctly.

- If axes are not marked, but intercepts are, deduct 1 point.
- If intercepts are not marked, but axes are, deduct 2 points.
- The coordinates at which the kink occurs should be marked clearly. If the shape of the graph is correct, but the coordinates for the kink are not marked, deduct 2 points.
- The analytical expression for the kinked budget line doesn't need to appear for full credits. However, it is useful to have it for the next part, so put it here as reference:

$$s = \begin{cases} 4.5 - 0.45c & 0 \leq c \leq 3, \\ 5.85 - 0.9c & 3 < c \leq 6.5. \end{cases}$$

- (b) (5 points) What quantity of coffee and pastries would Nazerke choose to consume given the new budget set? How does this answer compare with the previous result? Explain your answers.

The way to solve this problem is to guess that the solution will occur in the second part of the budget line, and then check that it is indeed the case. Thus, an optimal consumption bundle will satisfy:

$$\frac{3}{2} \frac{s}{c} = 0.9, \quad s = 5.85 - 0.9c,$$

assuming optimal coffee consumption will be larger than 3. Solving the two equations together yields:

$$c^M = 3.9, \quad s^M = 2.34.$$

Notice that the solution indeed satisfy our original assumption that coffee consumption would be greater than 3.

- Some version of the above procedure, together with the correct solution must appear for full credit.
- A maximum of two points can be awarded if they try some coherent procedure, but didn't reach the solution.
- Once again, students may have missed the assumption that we always make that fractional amounts are possible. If any student rounded up the coffee and pastries quantities, that is incorrect, because then the bundle is not feasible. If they rounded down any of them, deduct two points.

5. Suppose Daily cup decides to cancel its loyalty program

- (a) (4 points) Find Nazerke's consumption of coffee and pastries without any discounts.

$$c^M = \frac{3}{5} \times \frac{4500}{900} = 3, \quad s^M = \frac{2}{5} \times \frac{4500}{1000} = 1.8.$$

- This was just substituting values into the demand functions that they should have found previously in 2.b.
- In case they didn't find it, they should follow the standard procedure of equating the *MRS* the the appropriate price ratio of 0.9 and solve it together with the budget line.
- Full credits are for getting the right answer. A maximum of two points of partial credit can be awarded if they try some *coherent* procedure, but didn't reach the solution.
- Note that several students might have missed the traditional assumption that we always make in the problem sets and the lectures: *fractional amounts are possible*. Once again, students may have missed the assumption that we always make that fractional amounts

are possible. If any student rounded the consumption of pastries to 2 that is incorrect because it is not feasible. If they rounded it down to 1, deduct only one point if the procedure is correct.

- (b) (6 points) Comparing your answer to the one you got in 3.(b): how much of the change in Nazerke's consumption is due to the substitution effect? How much is due to the income effect? Perform the calculations and also illustrate your findings with a graph.

The original bundle, which we will call A , is $A = (6, 1.8)$, while the final bundle, which we will call B , is $B = (3, 1.8)$. To compute the compensated bundle, we need to solve the expenditure minimization problem. The conditions that identify the solution to the expenditure minimization problem are:

$$\frac{3}{2} \frac{s}{c} = 0.9, \quad c^3 s^2 = U(A) = 6^3 \times 1.8^2 = \frac{17496}{25} \approx 699.84.$$

Solving these two equations together yields the compensated bundle:

$$c^C = 3 \times 2^{3/5} \approx 4.54, \quad s^C = \frac{9 \times 2^{3/5}}{5} \approx 2.73.$$

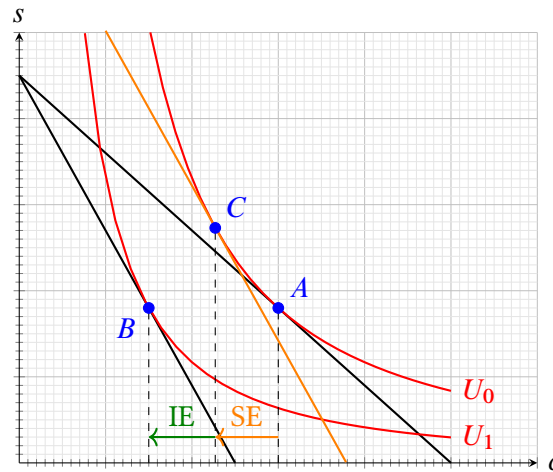
We denote the compensated bundle by $C = (4.54, 2.73)$.

Therefore:

$$SE = 4.54 - 6 = -1.46$$

$$IE = 3 - 4.54 = -1.54$$

$$TE = 3 - 6 = -3$$



- Full credits for this question involve some version of the steps above, the correct solution, and a moderately accurate graph, like the one above.
- If they managed to identify the compensated bundle, 3 points. Give partial credit depending on how far they got: 1 point for the correct optimality conditions, 2 points for solving it.
- Computing correctly (with the right sign) the income and substitution effects is 1 point each.

- The graph is 1 point, and it should resemble the one I drew above.
6. Nazerke is disappointed that the loyalty program has been canceled at Daily Cup. She decides to go to Health Project, where coffee sells for ₸800 Tenge and pastries for ₸800. Assume that the quality of coffee and pastries is the same between Health Project and Daily Cup.

(a) (4 points) How many coffees and pastries does Nazerke buy at Health Project?

$$c^M = \frac{3}{5} \times \frac{4500}{800} = 3.375, \quad s^M = \frac{2}{5} \times \frac{4500}{800} = 2.25$$

- Again, this was just substituting values into the demand functions they found previously.
 - In case they didn't find it, they should follow the standard procedure to find an optimal solution, equate the MRS to the price ratio of 1 in this case, and solve together with the budget line.
 - Full credits are for getting the right answer. A maximum of two points of partial credit can be awarded if they try some *coherent procedure*, but didn't reach the solution.
 - Once again, students may have missed the assumption that we always make that fractional amounts are possible. If any student rounded up the coffee and pastries quantities, that is incorrect, because then the bundle is not feasible. If they rounded down any of them, deduct two points.
- (b) (4 points) Nazerke only has time to go either Daily or HP before class. Which one does she go to? Why?

This is just a comparison of utilities: at Daily Cup Nazerke consumes the bundle $D = (3, 1.8)$, while at HP she consumes the bundle $H = (3.375, 2.25)$. The utilities of each bundle are:

$$U(D) = 87.48, \quad U(H) = 194.62.$$

Thus, Nazerke should go to HP.

- Full points are for the correct solution as above.
 - Two points can be awarded if they recognize that they just need to compare utilities. If they recognize this, and try to do the calculation, but they mess it up, give 3 points.
- (c) (4 points) Now suppose that Nazerke's utility function for coffee and pastries is given by $U(c, s) = \sqrt{c + s}$. Which coffee shop does Nazerke go and how much of each good does she consume at that coffee shop?

This is a perfect substitutes utility function, $x + y$, with the transformation $f(x) = \sqrt{x}$ applied. Thus, no major computation is needed: the absolute value of the marginal rate of substitution is $|MRS| = 1$. In Daily Cup, Nazerke will choose then to specialize in coffee because $1 > 0.9$, thus consuming the bundle $(5, 0)$. In HP, she is indifferent between any bundle that is in her budget line, i.e., any combination of the bundles $(5.86, 0)$ and $(0, 5.86)$. Therefore, Nazerke achieves a higher utility in HP.

- Full credits are for all parts of the argument as above.
- Recognizing that is perfect substitutes is 1 point.
- Solving for the Daily cup consumption is 1 point.
- Solving for the HP consumption is 1 point.
- The final observation that Nazerke prefers HP is 1 point.
- They don't need to do any more fancy calculations, but if they did, award points based on how the argument they did leads to the conclusion detailed above.